

Abstract

Container terminals schedule quay cranes and speed-adjustable automated guided vehicles against two conflicting objectives, the makespan of the workload and the energy consumed by the vehicles. The solution is not a single schedule but a Pareto front of non-dominated compromises from which an operator must select. The literature is substantially richer in methods that compute fronts than in methods that explain them, and the explainability techniques usually transferred from machine learning attribute outcomes to decision variables rather than to physical quantities on which an operator can act.

This thesis explains Pareto-optimal schedules in terms of the physical activities that produce them. The account rests on a structural observation: once a schedule is fixed, its makespan is the longest chain of activities that must happen in sequence, so only the activities on that chain can affect it, and their durations add up to the makespan exactly. The sensitivity of the makespan to any activity is therefore one or zero rather than something in between, and the resulting explanation balances by construction. Formally this is linearity over the $(\max, +)$ semiring, and it makes the explanation checkable rather than merely plausible. The method built on it is a heterogeneous graph attention network, E-HGATv2, which encodes the schedule as a typed precedence graph and predicts per-leg travel durations and per-task handling delays, which a differentiable max-plus dynamic-programming layer composes into the makespan, its custom backward pass routing the subgradient exclusively through the argmax edge of each recurrence.

From this decomposition the thesis defines the duration-weighted critical share ρ , the fraction of the makespan contributed by vehicle travel rather than crane handling, and uses its shift between the ends of a front to quantify why the compromises differ. The principal finding is that the binding bottleneck migrates between travel and handling, and that whether it does is governed by the fleet-to-crane ratio A/Q . Across 540 replicates spanning six instances, nine fleet sizes and ten seeds, migration is significantly non-zero in 45 of 54 instance–fleet cells under a Holm-corrected Wilcoxon signed-rank test, attaining 0.665 at $A/Q = 2.67$ and vanishing below $A/Q \approx 1$ where both ends are transport-bound. A second campaign of 8,640 replicates reproduces the threshold on three instance families and resolves the explanation at all 657,899 points of the fronts it produces. Where the transfer occurs, cranes bind at the makespan-optimal end and vehicles at the energy-optimal end, which accounts for its absence on the ten published large instances, each having $A/Q = 0.50$. The surrogate reproduces the exact explanation to a mean absolute error of 0.026 in the share and a critical-leg agreement of 0.898 on the small instances and 0.924 on the published large ones, and it does so at every generation of a search as well as at its end. That agreement is what carries the method beyond the cases examined here, since the exact recurrence can only be applied to a schedule that has already been evaluated whereas the model predicts the durations it needs.

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Chapter 1

Introduction

A container terminal mediates between maritime and inland transport, and its throughput is governed by the coordination of its equipment. Quay cranes transfer containers between vessel and quay; automated guided vehicles transport them between quay and storage yard. Scheduling the two resources jointly dominates scheduling them in sequence, because an idle crane represents unrecovered capital and an idle vehicle represents wasted energy. Contemporary terminals increasingly deploy vehicles whose travel speed is selectable, which introduces a second objective: a vehicle driven faster completes its leg sooner but consumes disproportionately more energy, since power grows super-linearly in speed.

The problem consequently admits no single optimum. It admits a Pareto front: a set of schedules, none of which improves upon another in one objective without conceding the other. Fontes and Homayouni [10] formulate the problem as a bi-objective mixed-integer linear program and solve it with a multi-population biased random-key genetic algorithm, reporting fronts for benchmark instances of increasing size. Their contribution concerns the efficient computation of the front.

The subsequent question has received less attention. An operator receives a front and must select from it. Selection presupposes understanding, and understanding a schedule requires knowing what limits it. Two schedules on the same front may be limited by entirely different resources: one may be constrained by vehicles traversing a long yard-to-quay distance, another by a single overloaded crane whose task queue no increase in fleet size will shorten. These situations require distinct operational responses, and the front alone does not distinguish them. The front reports the magnitude of the trade-off; it does not report its cause.

1.1 Significance

Two questions are worth separating: whether a front can be computed, and whether it can be understood. The first is settled for this problem; the second is what this thesis addresses, and it matters for a practical reason. A front is not a recommendation. It is a menu, and the operator who must choose from it is choosing between schedules that may fail for entirely different reasons. If one compromise is slow because too few vehicles are covering a long yard, buying a vehicle helps; if another is slow because a single crane has too many containers queued against it, buying a vehicle changes nothing and the tasks must be rebalanced. The two look alike on the front, because the front records only the makespan and the energy, never the reason. An operator told which resource is binding at the point they are considering can act; an operator given the two numbers alone can only guess.

The gap is not specific to container terminals. Bengio et al. [3] survey the integration of machine learning with combinatorial optimisation and identify the interpretation of optimiser output as a persistent weakness. Within multi-objective optimisation the usual devices are indirect: fronts are visualised, which conveys the shape of a trade-off but not its cause; solutions are clustered,

which groups compromises without saying what distinguishes them; or a preference is elicited and one solution recommended, which replaces one unanswered question with another. Recent work applies post-hoc attribution, principally SHAP [18] and TreeSHAP [19], to surrogate models of the objective. In a scheduling context that carries a specific limitation: the attribution lands on *decision variables*, on positions in a chromosome and on discrete speed settings, not on *physical quantities*. Being told that the twenty-seventh gene of a chromosome matters does not tell an operator what to do. Being told that the loaded trip from crane three to yard station one lies on the critical path and contributes 55.6 seconds does.

1.2 Novelty

What makes the problem tractable here is a structural property rather than a new technique. Rudin [21] argues that where a model’s structure permits an explanation that is correct by construction, explaining an opaque model after the fact is the inferior choice. This problem permits exactly that. The makespan of a fixed schedule is the largest total along any chain of activities that must happen in order, which is to say it is the length of a longest path through the schedule. Two consequences follow. Only the activities on that path can affect the makespan at all, so the sensitivity of the makespan to any activity is either one or zero, never something in between. And the durations of the activities on that path add up to the makespan exactly. The explanation is therefore not fitted or approximated; it is read off the structure, and it balances by construction. Formally this is linearity over the $(\max, +)$ semiring [2], developed in Section 2.3. The two ingredients this thesis combines are each established. Graph neural networks have been applied to scheduling and routing, as policies [17, 24] and as surrogates, and differentiable optimisation layers embed a combinatorial computation inside a network so that gradients pass through it [1, 20], with the decision-focused learning literature studying prediction whose loss is defined through a downstream optimiser [8]. Their combination for this problem class is, to the author’s knowledge, not established. The novelty is in what the network is asked to predict. It does not predict the makespan. It predicts the *local physical quantities* of a schedule, the duration of each vehicle trip and each crane lift, and a max-plus layer then composes those into the makespan using the same recurrence the exact evaluator uses. Differentiating the model with respect to its own predictions therefore returns the critical path itself rather than an estimate of it, and the answer is expressed in seconds attributable to named activities rather than in units of an internal representation.

1.3 Contribution

This thesis has one core idea: *a Pareto-optimal schedule can be explained exactly, in seconds attributable to named physical activities, and a learned model can produce that same explanation.* Everything reported here serves that idea.

The explanation itself is exact rather than estimated. For a given schedule the method identifies which vehicle trips and crane lifts actually determine the makespan, and how many seconds each contributes. Because those durations add up to the makespan exactly, the explanation can be checked rather than merely believed, and it is checked on every schedule reported, so an inconsistency appears as a failed test rather than as a plausible figure. This holds because

a fixed schedule’s makespan is the longest chain of activities that must run in sequence, which makes the sensitivity to any one activity either one or zero.

The learned surrogate, E-HGATv2, is the supporting tool. It is not asked to predict the makespan; it predicts the duration of each trip and each lift, and a differentiable max-plus layer composes those into the makespan using the same recurrence the exact evaluator uses. Differentiating the model with respect to its own predictions therefore returns a critical path rather than an estimate of one, expressed in the same physical units. Whether it agrees with the exact answer is measured throughout and reported in Section 4.7.

That the explanation is worth having at all is established by a single empirical fact: the answer is not the same everywhere on a front. With enough vehicles the makespan-optimal end of a front is limited by the cranes and the energy-optimal end by the vehicles, so two schedules an operator is choosing between fail for different reasons and call for different interventions. Whether this occurs is governed by the number of vehicles per crane, which is established by a statistically tested sweep (Section 4.4) and reproduced on three instance families. This is evidence for the core idea rather than a second contribution: without it, explaining a schedule would be a demonstration on one solution instead of a statement about the front.

Chapter 2 states the problem, its formulation, the solution method and the relevant background. Chapter 3 describes the surrogate architecture, the max-plus layer, the derived measure and the experimental design. Chapter 4 evaluates the results, beginning with verification. Chapter 5 states the limitations, including one degenerate case the implementation does not handle correctly. Chapter 6 concludes.

Problem Formulation and Background

2.1 The problem, its entities and the physics of a leg

An instance is a set of container moves (*tasks*) $\mathcal{J} = \{0, \dots, N-1\}$, each bound to one quay crane $qc(j)$ with handling time τ_j and one yard station $lu(j)$, served by a fleet of $|\mathcal{A}|$ speed-adjustable vehicles. A task is of one of two kinds. A *loading* task ($j \in \mathcal{L}$, a LOAD) moves a container from yard to vessel: a vehicle travels empty to the yard station, collects the container, carries it to the crane, and the crane lifts it aboard, so its loaded leg runs $LU \rightarrow QC$. An *unloading* task ($j \in \mathcal{U}$, an UNLOAD) reverses this, the crane lifting a container from the vessel onto a waiting vehicle which transports it to the yard. Each crane processes its tasks one at a time.

A decision specifies, for every task, the assigned vehicle, its position in that vehicle's sequence and in its crane's sequence, and a discrete *speed level* for the empty leg and for the loaded leg, one of a small fixed set of velocities at which a vehicle may be driven; a higher level shortens the leg and raises its power draw. The instances follow the three-level configuration of Fontes and Homayouni [10]. Two objectives are minimised: the makespan $C_{\max}$, the completion time of all work, and the energy E consumed by vehicle travel. Crane handling energy is excluded from E , which is material to the interpretation of both objectives, since the crane contributes to one and not the other.

Table 2.1 collects the notation. Sequencing is represented by the order of tasks within a resource's chain, and the formulation below states the problem as the timing recurrences a resolved schedule satisfies.

Table 2.1: Notation. The upper block is problem data, the middle block schedule variables, and the lower block quantities introduced by this thesis.

Symbol	Meaning
$\mathcal{J}, N$	set of container moves (tasks), $N = \mathcal{J} $
$qc(j), lu(j)$	the crane and yard station to which task j is bound
$qcprev(j)$	the predecessor of j in its crane's sequence
$Q, \mathcal{A} $	number of quay cranes and of vehicles
τ_j	quay-crane handling time of task j
α, v_ℓ, P_ℓ	speed multiplier, speed and power draw at level ℓ
$t_j^{\text{empty}}, t_j^{\text{loaded}}$	empty and loaded leg durations of task j
$e_j^{\text{empty}}, e_j^{\text{loaded}}$	empty and loaded leg energies of task j
a_j^{agv}, c_j	vehicle-ready time and completion of task j
$C_{\max}, E$	makespan and total vehicle energy
$\chi \in [0, 1]^{4N}$	random-key chromosome
G_s	heterogeneous schedule graph, arc types $E^{\text{agv}}, E^{\text{qc}}$
h_j	node embedding produced by the E-HGATv2 core
$w[v], w[u, v]$	node and edge weights of the max-plus dynamic programme
$g_a = \partial C_{\max} / \partial w_a$	attribution of activity a ; exactly 0 or 1
π^*	the critical (longest) path
$\rho_{\text{transport}}$	duration-weighted critical share, Eq. (3.2)
A/Q	fleet-to-crane ratio, the sweep parameter

Table 2.2: Kinematic and power constants per speed level, after Fontes and Homayouni [9]. These are the authors' own three-velocity scenario and are used unchanged.

Level	α	empty v (m/s)	empty P (kW)	loaded v (m/s)	loaded P (kW)
Lower	0.8	4.8	7.8	2.4	11.7
Nominal	1.0	6.0	10.0	3.0	15.0
Higher	1.2	7.2	13.2	3.6	19.8

This is the form the implementation computes and the form in which the max-plus structure of Section 2.3 is manifest. The mixed-integer programme of Fontes and Homayouni [10] formulates the same problem and supplies the published optima used for verification in Chapter 4.

Physics of a leg. Three speed levels $\alpha \in \{0.8, 1.0, 1.2\}$ scale a nominal speed; empty and loaded legs have distinct nominal speeds and power draws (Table 2.2). For a leg of length d travelled at level ℓ ,

$$t_{\text{leg}}(d, \ell) = \frac{d}{v_\ell}, \quad e_{\text{leg}}(d, \ell) = P_\ell \cdot t_{\text{leg}}(d, \ell), \quad (2.1)$$

so energy is power times time. A faster level shortens t but raises P , and this is the sole source of the time–energy trade-off.

2.2 Objectives and the completion recurrences

Total energy is additive over the chosen legs,

$$E = \sum_{j \in \mathcal{J}} (e_j^{\text{empty}} + e_j^{\text{loaded}}). \quad (2.2)$$

The makespan requires the precedence structure. Fixing the assignment and the sequencing turns the schedule into a precedence graph: task j cannot proceed until both its predecessor on the same vehicle and its predecessor on the same crane have finished. Writing a_j^{agv} for the time at which the vehicle assigned to j becomes free for it, and $\text{qcprev}(j)$ for j 's predecessor on its crane, the per-task completion obeys, for a LOAD,

$$c_j = \max(a_j^{\text{agv}} + t_j^{\text{empty}} + t_j^{\text{loaded}}, c_{\text{qcprev}(j)}) + \tau_j, \quad (2.3)$$

and for an UNLOAD,

$$c_j = \max(a_j^{\text{agv}} + t_j^{\text{empty}}, c_{\text{qcprev}(j)} + \tau_j) + t_j^{\text{loaded}}. \quad (2.4)$$

The makespan is then

$$C_{\max} = \max_{j \in \mathcal{J}} c_j. \quad (2.5)$$

The asymmetry between (2.3) and (2.4) is an essential feature. For a LOAD the handling τ_j is added after the maximum, because the crane must lift the container once the vehicle has delivered it; for an UNLOAD the handling appears within the maximum on the crane branch only, because the crane's lift overlaps the vehicle's travel to the quay, and the loaded leg is then added afterwards. A LOAD therefore completes when its crane finishes, whereas an UNLOAD completes when its vehicle reaches the yard. Chapter 4 shows that this asymmetry

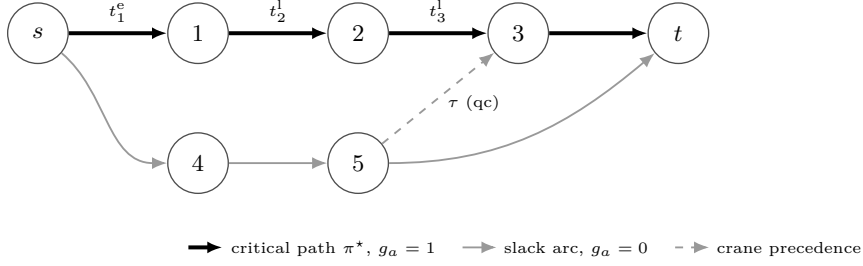


Figure 2.1: The event DAG of a small schedule. Solid arcs are vehicle legs and the dashed arc a quay-crane serialisation constraint. The bold path $s \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow t$ is the critical path π^* ; by Section 2.3 only its arcs carry a non-zero attribution, and their durations sum to $C_{\max}$ by Eq. (2.6).

is precisely what reconciles the single-maximum objective with the two-array formulation used in the reference implementation.

2.3 The makespan as a longest path

Each constraint above asserts that a completion time is at least the sum of a predecessor and a duration. At optimality every such variable equals the greatest of its lower bounds. Writing $\oplus$ for maximum and $\otimes$ for addition, the schedule is a linear system over the $(\max, +)$ semiring [2, 13], and $C_{\max}$ is the length of a longest path in the directed acyclic graph whose vertices are activities (empty legs, loaded legs and crane handlings) and whose arcs are the precedences above. The construction is the disjunctive graph familiar from job-shop scheduling [5], specialised to two resource types.

Two consequences follow, and this thesis rests upon both.

First, the objective depends only on the activities of one path. Perturbing the duration of an on-path activity changes $C_{\max}$ by the same amount; perturbing an off-path activity leaves $C_{\max}$ unchanged until the perturbation is large enough to alter the longest path. The sensitivity is thus piecewise constant, equal to one on the critical path and zero elsewhere.

Second, as an immediate corollary, the durations of the on-path activities sum to the makespan:

$$C_{\max} = \sum_{\text{on-path legs } \ell} d_{\ell} + \sum_{\text{on-path tasks } j} \tau_j. \quad (2.6)$$

This is an identity, not an approximation, and it is the property that the architecture of Chapter 3 is designed to preserve. Figure 2.1 illustrates the construction on a small schedule.

2.4 The multi-population biased random-key genetic algorithm

Fronts are produced by the mp-BRKGa of Fontes and Homayouni [10], reimplemented against the authors' own solver source. The reimplementaion was diffed against that source operator by operator before any result was produced, and the correspondence is stated exactly rather than assumed. The objective is identical, not approximately so: his is split across a per-task completion array and a per-vehicle availability array, ours takes a single maximum over completions, and the asymmetry of (2.3) and (2.4) makes the two equal, which Chapter 4 confirms on 2400 random schedules at exactly zero difference. The elite and mutant counts use his `ceil` rounding, and the second crossover parent is drawn from the non-elites as in his disjoint parent ranges. One representational difference remains: his mutants store integer speed and vehicle

indices directly in blocks two to four, whereas ours stores random keys that the decoder buckets into the same levels. The induced distribution over decoded mutants is therefore the same, and was checked to be so, but the chromosome encoding differs and a seed-for-seed reproduction of his trajectory is not possible. Following the biased random-key paradigm [12] a solution is a vector of $4N$ keys, one block of task priorities, two of empty- and loaded-leg speed levels and one of vehicle assignments, which a decoder maps to a schedule that is then evaluated exactly. The method evolves $\Omega + \Pi$ populations concurrently: each single-objective population ranks by one objective and so drives one extreme of the front, while each multi-objective population ranks by non-dominated sorting with crowding distance [7]. Each generation replaces a population by its elite fraction, offspring from biased parameterised uniform crossover inheriting from the elite parent with probability $p_e = 0.7$, and fresh mutants; elites migrate between populations so that boundary specialists propagate inward.

The published configuration is $P = 20N$ per population, elite fraction 0.2, mutant fraction 0.1, $p_e = 0.7$, $\Omega = \Pi = 2$ and $n_{ex} = 30$, adopted here with the reduced generation budget of Section 3.6. The search is not a contribution of this work; it is the instrument supplying Pareto-optimal schedules to be explained, and no learned model participates in producing any front reported here.

2.5 Selection of the surrogate model class

Exact evaluation costs well under a millisecond at these sizes, so a surrogate is not required to render search tractable. Its purpose is that a model predicting the per-activity structure of a schedule can be differentiated with respect to its own predictions, yielding an explanation derived from a learned representation rather than from the exact recurrence, which the two can then be compared on. Establishing that the explanation is learnable is a precondition for any setting in which the exact recurrence is unavailable.

Three properties of the input determine the model class. First, a resolved schedule is a typed precedence graph, not a vector. Any model consuming a fixed-length feature vector must serialise it as the $4N$ chromosome of Section 2.4, as a tree ensemble such as XGBoost [6] would; that representation has a length depending on N , so it does not transfer across instance sizes, and its coordinates are decision-variable indices, so any attribution over it, TreeSHAP [19] included, attributes to chromosome positions rather than to physical activities. Second, the recurrences (2.3)–(2.4) relate a task to its immediate predecessor on a resource, which is message passing [11]: a graph network is constrained to the locality a multilayer perceptron would have to learn. Third, the two relations enter (2.3) and (2.4) differently and therefore cannot share parameters, and the quantities that matter, travel time and leg energy, are properties of arcs rather than nodes, which forces edge conditioning.

These jointly specify a heterogeneous, edge-conditioned message-passing network. The attentional variant is adopted in the corrected GATv2 form of Brody et al. [4] rather than the original GAT [22], whose attention is monotonic in the key and therefore cannot rank neighbours conditioned on the query. Cross-relation fusion follows the semantic attention of Wang et al. [23]; Section 3.3 explains why that cross-relation attention, and not the within-relation attention, carries the informative signal here.

Method and Experimental Design

3.1 Research question

The investigation asks whether, for a schedule on the Pareto front of the problem of Chapter 2, one can state exactly why it is efficient, and whether that explanation varies systematically along the front. The first part is addressed constructively by the architecture below; the second is empirical and requires a scalar summary of an explanation.

3.2 Graph representation

Figure 3.1 gives the pipeline as a whole; this subsection and the two that follow describe its three stages.

A resolved schedule is encoded as a heterogeneous graph. Nodes are tasks, carrying four features: the handling time τ_j , two indicators distinguishing loading from unloading, and the identifier of the servicing crane. Arcs are of two types. An *AGV arc* connects consecutive tasks on the same vehicle; a *QC arc* connects consecutive tasks on the same crane. Each arc carries three features: the total travel time of the leg it represents and the empty and loaded energies of that leg.

The graph is therefore precisely the precedence DAG of Figure 2.1, so the representation is isomorphic to the combinatorial structure of the objective. Two aspects are consequential: each task has at most one incoming arc of each type, with the architectural implication of Section 3.3, and travel time and energy reside on arcs, so the model must condition its messages on edge attributes rather than on node states alone.

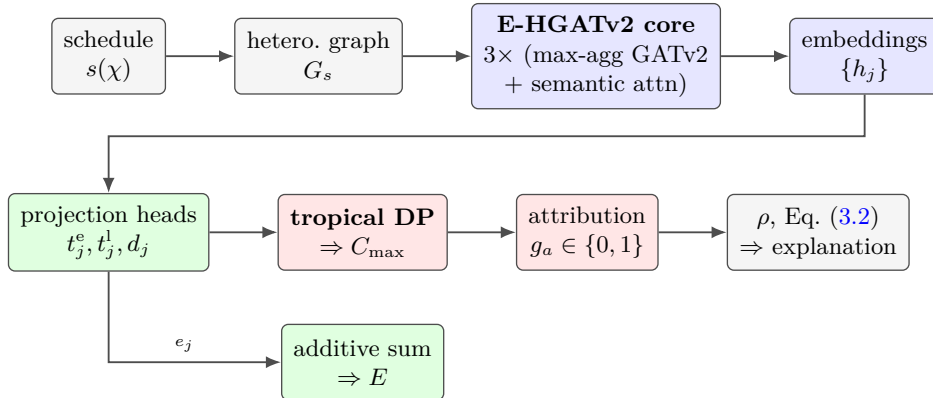


Figure 3.1: The pipeline. The core embeds a decoded schedule; projection heads predict *local physical attributes*, which an exact max-plus dynamic programme composes into $C_{\max}$, energy being an exact additive sum. Differentiating the dynamic programme yields the binary attributions $g_a = \partial C_{\max} / \partial w_a$, from which ρ is computed. The network never regresses an objective directly.

3.3 The E-HGATv2 architecture

Each layer updates the fixed-length description held at every task by combining it with those held at its immediate predecessors on its vehicle and on its crane, so stacking L layers lets information reach a task from predecessors up to L positions away in either chain. The surrogate has three such layers, a hidden dimension of 64 and four attention heads per layer, and each layer performs two operations.

Within-relation propagation under a max-plus reduction. For each relation $R \in \{\text{AGV}, \text{QC}\}$ separately, a GATv2 convolution [4] conditioned on the arc attributes computes a message for every node from its predecessors under R . The aggregation is $\max$ rather than the customary attention-weighted sum. This choice mirrors the semiring: the recurrences of Section 2.3 reduce a node’s same-resource predecessors under a maximum, and the aggregator is therefore selected to match the algebra of the problem rather than for its smoothness.

Cross-relation fusion by semantic attention. The two relation messages are combined by a learned attention over relations, in the manner of Wang et al. [23]. This is where the informative signal resides, and the reason is a direct consequence of the encoding of Section 3.2. Since a resolved schedule gives each task at most one predecessor per resource, the within-relation attention softmax is computed over a single incoming arc and is therefore degenerate, equal to one identically. It contributes no discriminative information. The non-degenerate quantity is which of the two resources gates the task, exactly the $\max(\text{agv_ready}, \text{qc_ready})$ argmax of the recurrence, and this is what the cross-relation attention represents. Its per-node weight on a relation is interpretable as the criticality of the corresponding incoming arc, and it is recorded per arc for that purpose.

This asymmetry inverts the conventional reading of an attentional graph network, in which the within-neighbourhood attention constitutes the object of interest. In the present setting that component is structurally vacuous, and the cross-relation component carries the resource-contention signal. Figure 3.2 shows one layer.

3.4 The tropical max-plus layer

The architecture described so far could be completed in the conventional manner, by pooling node embeddings and regressing $(C_{\max}, E)$ with a multilayer perceptron. Doing so defeats the purpose of the exercise. A pooled regression head mixes every node embedding into every output, so the Jacobian of the predicted makespan with respect to the input features is dense. An attribution read from it distributes credit across activities that are provably irrelevant, because Section 2.3 establishes that only the activities of one path can affect $C_{\max}$ at all. The head, not the encoder, is what destroys faithfulness.

The makespan head is therefore replaced by a differentiable max-plus dynamic programme. The layer computes, for a directed acyclic graph of activity events,

$$x[v] = w[v] + \max_{u \rightarrow v} (x[u] + w[u, v]), \quad (3.1)$$

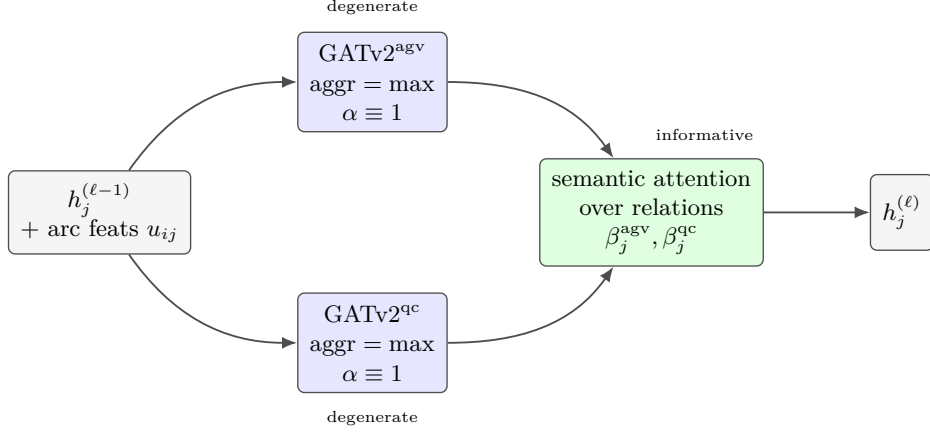


Figure 3.2: One heterogeneous layer. Message passing runs independently per relation under a maximum aggregation, and the two relation messages are then fused by a learned attention across relations. Within a relation a resolved schedule gives each task at most one incoming arc, so the GATv2 softmax is over a single arc and is degenerate ($\alpha \equiv 1$). Across relations β_j is non-degenerate and encodes which resource gates task j , the $\max(\text{agv_ready}, \text{qc_ready})$ argmax of the recurrence; it is recorded per arc as that arc’s criticality. The informative attention is therefore the cross-relation one.

in a deterministic topological order obtained by Kahn’s algorithm, with node weights $w[v]$ and edge weights $w[u, v]$ supplied by the network. The makespan is the maximum over sinks. The layer is implemented as a custom autograd operation: the forward pass records the maximising predecessor of each node, and the backward pass routes the incoming subgradient exclusively through that recorded argmax edge.

The consequence is the property sought. Because the backward pass follows a single path, $\partial C_{\max}/\partial(\text{activity duration})$ is exactly one for activities on the critical path and exactly zero otherwise. Attribution is not estimated; it is read off a selection that the forward pass already made. Figure 3.3 contrasts this with the pooled head it replaces.

Equation (2.6) then holds for the model’s own predicted durations exactly as it holds for the exact ones, which is what permits the model’s explanation and the exact explanation to be compared on identical terms. The layer belongs to the family of differentiable optimisation layers [1, 20], and is distinguished by the fact that the embedded computation is a dynamic programme whose subgradient is available in closed form rather than an optimisation requiring implicit differentiation.

Ties in (3.1) are resolved by the first maximum in edge order, following the deterministic **argmax** convention. This is a valid subgradient selection, since any single maximiser of a maximum is a subgradient; Chapter 5 reports a defect arising where the surrounding analysis code does not respect this convention.

Prediction targets, anchoring and the energy head. The network does not predict the objectives. An arc-level head maps the concatenated endpoint embeddings of a task’s incoming AGV arc to its empty and loaded travel times and leg energies, and a node-level head maps a task embedding to its crane handling delay; the makespan is composed from these by (3.1) and the energy by exact summation, a linear head for which $\partial E/\partial e_\ell = 1$ identically. Because a max-plus output constrains only the attributes on the currently active path, training on the makespan alone would leave off-path attributes unidentified, so the local attributes are supervised directly

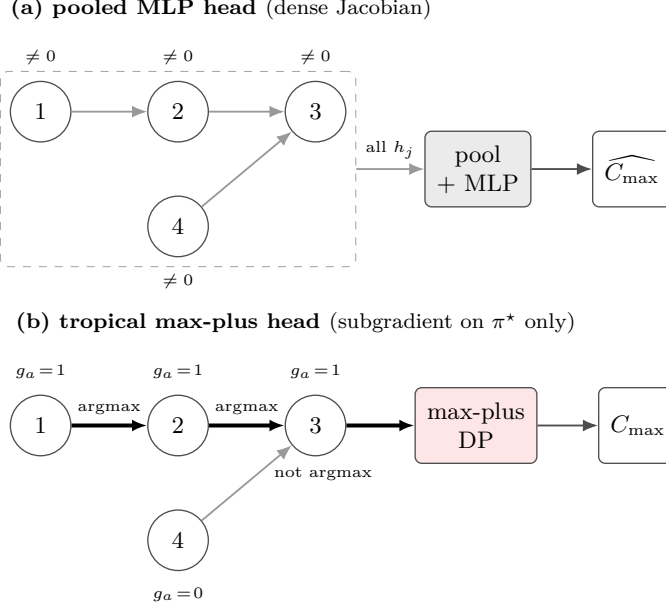


Figure 3.3: The output head, not the encoder, determines whether the attribution is faithful. Both panels encode the same four activities, three on the longest path while activity 4 loses the maximum at node 3. (a) A pooled regression head mixes every node embedding into the output, so every activity receives a non-zero derivative, including activity 4, which provably cannot affect $C_{\max}$. (b) The max-plus layer records the maximising predecessor during the forward pass and routes the subgradient through it alone, so the attribution is the indicator of the critical path.

against their exact physical values. Training is therefore two-stage rather than end-to-end: the core is trained first, then frozen with cached embeddings while the heads are fitted. One consequence should be read with care. The arc features the encoder aggregates include the per-leg energies themselves, so predicting total energy is close to reading a sum of inputs, and the near-unit coefficient of determination in Section 4.7 is a property of the architecture rather than evidence of learning.

3.5 The duration-weighted critical share

Comparing explanations across schedules requires a scalar. The natural candidate is the proportion of the critical path attributable to vehicle travel rather than crane handling.

Counting on-path activities of each kind is inadequate. Legs and handlings are not commensurable: a task may contribute a critical handling while its vehicle chain also binds, so the two counts neither partition the path nor sum to a meaningful total. An earlier formulation of this work used counting and produced a measure that failed to reconcile with the makespan.

The measure adopted here weights by duration. Define

$$\rho_{\text{transport}} = \frac{\sum_{\text{on-path legs}} d_{\ell}}{\sum_{\text{on-path legs}} d_{\ell} + \sum_{\text{on-path tasks}} \tau_j}, \quad \rho_{\text{handling}} = 1 - \rho_{\text{transport}}. \quad (3.2)$$

By (2.6) the denominator equals $C_{\max}$ exactly, so the two shares partition the makespan and sum to unity by construction. Neither term counts objects, which removes the commensurability problem entirely.

The quantity of interest is the shift in $\rho_{\text{transport}}$ between the two ends of the front,

$$\text{migration} = \rho_{\text{transport}}(\text{makespan-optimal}) - \rho_{\text{transport}}(\text{energy-optimal}). \quad (3.3)$$

The sign is retained rather than taken in absolute value, because it identifies which end is travel-heavy. A negative value indicates that the energy-optimal end leans more on travel than the makespan-optimal end does, so that prioritising energy moves the bottleneck towards vehicle travel; a positive value indicates the converse; and a value near zero indicates that the character of the bottleneck is invariant along the front even though the objective values are not.

3.6 Experimental design

Instances. Two published sets are used directly [9, 10], and a further two families are constructed from the first as Section 4.5 describes. The small set comprises 35 loading instances of 4 to 16 tasks on a six-crane, six-station layout. The large dual-cycling set comprises 10 instances of 40 to 160 tasks on a twenty-crane, twenty-station layout; it is referred to throughout as the large dual-cycling set, to distinguish it from the 26 small mixed instances of Section 4.5, which are dual-cycling in the same operational sense but are a different set on a different layout. Both were transcribed programmatically from the published tables and the authors’ solver sources; no value was retyped. The transcription was cross-validated: the distance matrix and task lists were parsed independently twice and agreed in every cell, and four published optima and five mixed-instance pairings matched exactly.

Fleet sweep. The parameter of interest is the fleet-to-crane ratio A/Q . Because every published large instance has $A/Q = 0.50$, the published sets cannot by themselves determine whether migration depends on this ratio. Six small instances (L01, L07, L15, L21, L28, L35) were therefore each run at nine fleet sizes, $A \in \{1, \dots, 9\}$, with ten seeds: 540 replicates, all other parameters fixed.

Front campaign. The sweep above varies the ratio on six loading instances and records the transport share at the two extremes of each front. A second campaign widens both axes. It covers all 96 instances of the three families of Section 4.5 at the same nine fleet sizes and ten seeds, giving 8,640 replicates, and computes the share at every point of every front rather than at its extremes, giving 657,899 explained schedules. No surrogate is fitted in this campaign: every quantity is produced by the exact evaluator and the exact attribution, so a replicate costs a few seconds and the whole campaign runs in about ten minutes on 200 cores.

Budget and statistics. Each replicate runs mp-BRKGA at $P = 20N$ per population with $\Omega = \Pi = 2$, $n_{ex} = 30$ and 100 generations, below the published 300; since no comparison is against a published front and every arm shares the budget, the reduction affects absolute front quality but not the migration measurement. Migration is tested against zero by a one-sample Wilcoxon signed-rank test across the ten seeds of each instance–fleet cell, with p -values Holm-corrected [14] within each instance’s nine fleet sizes so that nine simultaneous tests cannot produce a spurious rejection. Effect size is the matched-pairs rank-biserial correlation [16]. A

non-parametric procedure is used because ten observations per cell do not support a distributional assumption and several cells are degenerate with near-zero variance. Front quality, where reported, uses hypervolume [25] and IGD^+ [15].

3.7 Software architecture and implementation

The implementation is a Python package of 50 modules and approximately 10,200 source lines, organised so that the exact physics, the learned model and the search are separable and independently testable. The `environment` package is the exact physics and carries no learned component; `surrogate` holds the graph construction and the encoder; `explain` holds the tropical layer, the fused head and both explainers; and `baselines`, `metrics` and `benchmark` form the evaluation harness.

The exact evaluator as label source. There is no external dataset. Every training target is produced by the exact evaluator, which runs (2.3)–(2.5) in topological order in $O(N)$ and returns both the objective pair and the per-task quantities the projection heads are supervised on. Random-key vectors are sampled uniformly, decoded and evaluated, so the data is generated self-supervised. This is what makes the physics anchoring of Section 3.4 possible, and it is why the evaluator’s agreement with the source work is a precondition for every other result.

The autograd function. The max-plus layer is a `torch.autograd.Function` rather than a composition of differentiable primitives, because the required backward behaviour is not the one automatic differentiation would supply for a chain of `torch.maximum` calls. The forward pass establishes a deterministic topological order by Kahn’s algorithm, rejecting any input that is not acyclic, since the precedence graph of a feasible schedule must be a DAG. It then relaxes (3.1) in that order, and stores the maximising predecessor of every node. The backward pass then walks the stored selections in reverse, accumulating the incoming gradient into exactly one predecessor per node. The `argmax` used in the forward pass is what implements the tie convention: it returns the first maximal index, so a tie yields one valid subgradient rather than a blend of several.

Chapter 5 records a case in which downstream analysis code violated this convention and the decomposition consequently failed to close.

Assertions, testing and execution. Node and arc feature tensors are validated at every entry point against declared semantic signatures, and the decomposition identity (2.6) is asserted on every traversal at a relative tolerance of 10^{-6} , so an inconsistency raises rather than being averaged into a table. The suite comprises 29 unit-test modules and 239 tests, all passing, including the equivalence of the max-plus evaluator with an exhaustive enumeration of all 1248 schedule structures on a five-task instance and the mp-BRKGA operator semantics against the authors’ source. Campaigns ran on a rented cloud instance of 384 vCPU and 125 GB RAM under Python 3.12 and PyTorch 2.11, distributed over a process pool; the 540-replicate sweep took 47 minutes and the 350-replicate small-set campaign 10 minutes, neither with a single failure.

3.8 Reproducibility

Every table in Chapter 4 is generated programmatically from the stored experiment output, so no figure in this document is transcribed by hand. Replicates are seeded and isolated: a failure in one cannot affect another, and failures are counted in the output. Every replicate of every campaign ran to completion: no run crashed, timed out or was lost, over the 940 replicates of the sweeps and the 8,640 of the front campaign alike. The decomposition check of Chapter 5 is satisfied as widely: the recovered critical path reconciles with the makespan on all 940 replicates and at all 657,899 front points, so no quantity reported in this document is computed on a filtered subset. An earlier version of the attribution failed that check on a small fraction of cases; the defect, its diagnosis and its correction are reported in Chapter 5.

Results

4.1 Verification against the source work

Before any explanation may be trusted the evaluator must agree with the source work. Fontes and Homayouni [10] compute the objective as $\max(\max_j c_j, \max_a A_a)$, in which A_a tracks when vehicle a next becomes available, whereas the implementation here takes a single maximum over per-task completions. The expressions are identical, and the reason is the asymmetry of (2.3) and (2.4): for a LOAD the availability equals $c_j - \tau_j$, the vehicle being released as soon as the crane assumes the container, and is dominated by $\max_j c_j$; for an UNLOAD it equals the completion of the loaded leg, which by (2.4) is exactly c_j . The reference implementation distributes across two arrays what (2.5) folds into one. On 2400 random schedules across eight instances of both sets the two expressions agreed to exactly zero difference in every case.

A second, independent check compares the makespan-optimal end of the computed fronts against the mixed-integer optima published for all 35 small instances [9]. This is not a test of exact agreement: the published optima are computed for fixed-speed scenarios, whereas the present runs select among three speeds per leg and therefore solve a less constrained problem whose makespan should lie at or below any fixed-speed optimum. A value above one would indicate an evaluator error or a search shortfall; a value below is the expected consequence of the enlarged decision space.

Table 4.1 reports the comparison. Across all four published makespan columns the best computed makespan lies at or below the published value in effectively every comparable cell, the two exceptions being L03 and L25, which exceed the nominal-higher column by 0.56 and 1.56 seconds respectively.

The loading family is Table 5 as printed, but the unloading and mixed families are reconstructions, so each is checked against the optima published for it. Table 4.2 reports both.

The closest correspondence is with the normal-higher scenario, at a mean gap of -17.26 seconds, and on L01 the agreement is exact: the makespan-optimal end is 175.00 against a published 175. That this is the nearest column is the expected outcome, since that scenario permits the

Table 4.1: The makespan-optimal end of the computed fronts against the published fixed-speed optima, over the 35 small instances. The present runs select among three speeds per leg and therefore solve a less constrained problem, so the computed value should lie at or below each published optimum; a value above one would indicate an evaluator error or a search shortfall. The nominal-higher scenario is the nearest comparison because it permits the highest fixed speed. 14 published cells are blank, the authors’ solver having failed to close them, and are skipped rather than imputed.

Published column	comparable	at or below	exact	mean gap (s)
nominal, $C_{\max}$	35	35	0	-365.75
nominal, $C_{\max}^*$	35	35	0	-98.69
lower-nominal, $C_{\max}$	33	33	0	-530.13
nominal-higher, $C_{\max}^*$	23	21	1	-17.26

Table 4.2: The two reconstructed instance families against the mixed-integer optima published for them, at the published fleet size of two vehicles. The unloading family reverses the origins and destinations of Table 5; the mixed family combines the pairs named in the “comb. of” column of Table 3, and is the 26-instance small set, not the ten large dual-cycling instances of Table 4.4. Ratios below one are expected, since the runs here select among three speeds per leg where the published optima fix one, so they solve a less constrained problem. The final column counts instances whose energy optimum is reproduced exactly, which the speed selection cannot improve upon and which therefore tests the reconstruction directly.

Family	instances	$C_{\max}$ ratio		E ratio	E exact
		mean	range	mean	
unloading	35	0.915	0.740–0.975	0.996	18/33
mixed	26	0.935	0.793–1.030	0.960	11/25

Table 4.3: Worked example: the exact makespan critical path of the makespan-optimal Pareto schedule of L01 ($N = 4$, $A = 2$, $Q = 2$). The on-path durations sum to $C_{\max} = 175.000$ exactly, because every on-path activity carries a unit subgradient. No surrogate is involved in producing this table.

Order	Task	Activity	Duration (s)	$\partial C_{\max}$	Resource
0	t_2	empty_leg	0.000	1	AGV 1, QC2
1	t_2	loaded_leg	55.556	1	AGV 1, QC2
2	t_1	empty_leg	27.778	1	AGV 1, QC1
3	t_1	loaded_leg	41.667	1	AGV 1, QC1
4	t_1	qc_handling	50.000	1	AGV 1, QC1
Sum of on-path durations			175.000		$= C_{\max}$
of which travel / handling			125.000 / 50.000		$\rho = 0.714$

highest fixed speed and is therefore the fixed-speed problem closest to the present one. The check establishes that the evaluator operates in the correct regime and never produces an implausibly low makespan against an independently computed bound; it does not, and cannot, reproduce the published values.

4.2 A worked example

Table 4.3 presents the complete exact explanation of a single schedule: the makespan-optimal Pareto point of L01, the smallest instance and therefore verifiable by hand. The critical path comprises five activities whose durations sum to the makespan exactly, as (2.6) requires.

Two properties of this traversal are of note. Every on-path activity belongs to the same vehicle, which establishes that the schedule is vehicle-limited and that the pertinent intervention is fleet capacity rather than crane rebalancing. The division between travel and handling is quantified rather than asserted: $\rho = 0.714$ indicates that slightly over seventy per cent of the makespan consists of vehicle movement and the remainder of crane handling. This constitutes the form of statement the thesis set out to produce.

4.3 The degenerate regime

Table 4.4 applies the method to both front extremes of all ten large dual-cycling instances, at up to 160 tasks. The decomposition closes on all 50 replicates.

Migration is near-absent throughout: mean absolute migration is 0.031, and the transport share lies between 0.87 and 0.97 at both ends of every front. Considered in isolation this constitutes a negative result for the hypothesis that the bottleneck migrates.

The explanation lies in the third column. Every one of the ten published large instances has

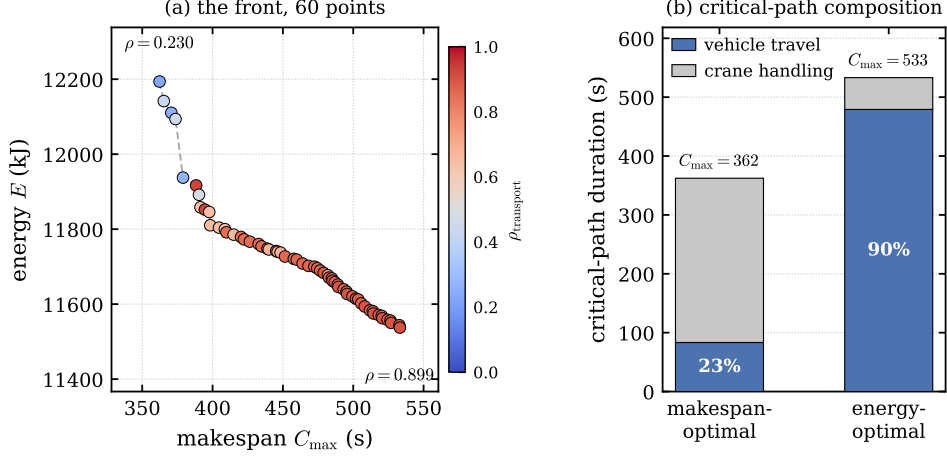


Figure 4.1: One explained front, for L21 at $A/Q = 1.67$ and seed 0, whose ten-seed cell mean appears in Table 4.5. (a) All sixty non-dominated points on the objective plane, each shaded by its duration-weighted transport share, so that the figure reports not only the trade-off but what binds at each compromise. The makespan-optimal end is crane-bound ($\rho_{\text{transport}} = 0.230$) and the energy-optimal end vehicle-bound ($\rho_{\text{transport}} = 0.899$); the transition between them occurs across the front rather than at its ends. (b) The critical path of each extreme resolved into its travel and handling parts, which sum to that end’s makespan by Eq. (2.6). The share, not the objective value, is what the explanation reports.

Table 4.4: Exact explanation of both front extremes on the ten large dual-cycling instances, averaged over five seeds each. Every instance has $A/Q = 0.50$, and the transport share is high at both ends, so migration is negligible throughout. Migration is the makespan-optimal share minus the energy-optimal share, as in Table 4.5. This is the degenerate regime that motivates the sweep in Table 4.5. The final two columns are the learned surrogate’s own migration on the same fronts and the Jaccard agreement of its recovered critical leg set with the exact one, so the model is measured on the largest instances in the study rather than only on the small ones. The decomposition closes on all 50 replicates, so none is excluded.

Instance	N	A/Q	n	exact		exact share		surrogate	
				$C_{\max}^{\text{mk}}$	$C_{\max}^{\text{en}}$	ρ_{mk}	ρ_{en}	migration	leg J
DL01	40	0.50	5	1799.8	3633.8	0.963	0.874	+0.089	0.960
DL03	50	0.50	5	2311.3	3754.3	0.971	0.890	+0.100	0.909
DL05	60	0.50	5	3183.3	5280.0	0.967	0.948	+0.019	0.908
DL07	70	0.50	5	3765.7	6121.1	0.959	0.909	+0.046	0.917
DL02	80	0.50	5	4178.1	6516.3	0.941	0.905	+0.036	0.966
DL09	80	0.50	5	3916.9	5569.6	0.959	0.954	+0.023	0.868
DL04	100	0.50	5	5364.7	7897.7	0.951	0.929	+0.025	0.920
DL06	120	0.50	5	6977.5	10763.8	0.955	0.955	−0.000	0.927
DL08	140	0.50	5	8377.2	12595.0	0.956	0.951	−0.000	0.944
DL10	160	0.50	5	8511.5	13456.2	0.972	0.968	+0.005	0.924
Mean migration +0.031, mean migration 0.031									

$A/Q = 0.50$ exactly, twice as many cranes as vehicles. With vehicles thus scarce, the schedule is transport-bound wherever it lies on the front: slowing it to conserve energy slows the vehicles, which already constituted the binding resource, so no transfer occurs. There is no bottleneck to migrate because only one candidate was ever available. This motivates the sweep rather than excusing the result, and it yields a testable prediction: migration should appear once vehicles are sufficiently numerous for cranes to become competitive.

Figure 4.2 presents both components of this account on the published set. Panel (a) reports the two ends of every DL front. The transport share exceeds 0.87 at the energy-optimal end and 0.94 at the makespan-optimal end of all ten instances, so the vehicles bind at either extreme and the separation between the two markers, which is the migration, is correspondingly small. Panel (b) places the set beside the small instances. At the common ratio the two sets are statistically

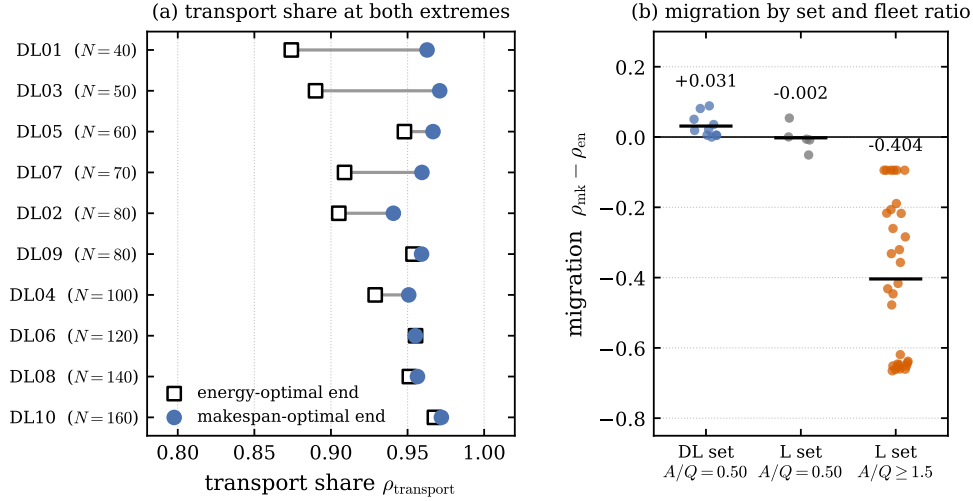


Figure 4.2: The published large dual-cycling set, over five seeds per instance. (a) Transport share at the two ends of every DL front, ordered by instance size. Both ends lie above 0.87, and the separation between them is the migration. (b) Migration for the DL set at its native $A/Q = 0.50$, for the small instances at the same ratio, and for the small instances at $A/Q \geq 1.5$, with group means marked. The first two groups are comparable in magnitude and the third is not. The decomposition closes on all 50 DL replicates.

indistinguishable, at +0.031 and -0.002 respectively, against -0.404 for the small instances at higher ratios. The published set is therefore not a family in which migration fails to occur, but a family observed at a single fleet ratio that lies below the threshold.

4.4 Bottleneck migration across the fleet-to-crane ratio

Table 4.5 reports the sweep and Figure 4.3 plots it, one panel per instance. Migration differs significantly from zero in 45 of the 54 instance–fleet cells after Holm correction, and the pattern is consistent across instances: negligible at low ratios, rising sharply near unity, and saturating thereafter.

Forty-one of the 45 significant cells carry a Holm-corrected p of 0.0176, the smallest value a ten-sample Wilcoxon test can attain, with a rank-biserial effect size of -1.00 in all but seven cells; the test therefore separates the two regimes without discriminating within either. At the lowest ratios migration is negligible. L07 at $A/Q = 0.50$ yields +0.000 and is one of the nine cells failing significance, every one of which lies at $A/Q \leq 1.17$, and L15, L28 and L35 all yield values within 0.06 of zero below $A/Q \approx 1$. This reproduces, from independent runs, the regime of Section 4.3 at the same ratio, which is the confirmation that section predicted.

Migration then increases sharply and saturates. L07 moves from -0.088 at $A/Q = 1.00$ to -0.660 at 1.50, thereafter remaining near -0.64 for every larger fleet. L21 attains -0.658 by $A/Q = 1.00$ and holds within 0.02 of -0.66 across the remaining six ratios, peaking at -0.665 . L28 crosses at $A/Q = 1.25$ and L35 at 1.33, both later than L07 and L21. The threshold is therefore instance-dependent rather than universal, which is to be expected: whether cranes can become binding depends on handling times and layout geometry, not on the ratio alone.

The sign identifies the direction of the transfer. Where migration is substantial it is negative, so by Eq. (3.3) the energy-optimal end is the travel-heavy one and the bottleneck moves towards vehicle travel as energy is prioritised. The worked case of Figure 4.1 makes this concrete on a single replicate of that cell: at the makespan-optimal end of L21 at $A/Q = 1.67$ the critical path

Table 4.5: Bottleneck migration across the fleet size, for the six swept instances. Migration is the signed shift of the duration-weighted transport share, $\rho_{mk} - \rho_{en}$, on the exact critical path, each cell the mean over ten seeds. Beneath each cell the model’s own migration is given, and beneath that the ratio A/Q followed by the mean absolute difference |

Δ

ρ between the two. The ratio differs between instances because the crane count does. Parenthesised cells are those not significant at $\alpha = 0.05$ under a one-sample Wilcoxon signed-rank test against zero, Holm-corrected within each instance’s nine fleet sizes. Migration is negligible left of $A/Q \approx 1$ and saturates to its right.

Instance	fleet size A , with A/Q beneath								
	1	2	3	4	5	6	7	8	9
L01	−0.051	+0.108	−0.432	−0.094	−0.094	−0.094	−0.094	−0.094	−0.094
model	−0.051	+0.100	−0.432	−0.094	−0.095	−0.095	−0.095	−0.095	−0.096
	0.50, 0.001	1.00, 0.012	1.50, 0.001	2.00, 0.001	2.50, 0.001	3.00, 0.001	3.50, 0.001	4.00, 0.001	4.50, 0.001
L07	(+0.000)	−0.088	−0.660	−0.648	−0.654	−0.651	−0.638	−0.643	−0.619
model	−0.000	−0.104	−0.606	−0.648	−0.611	−0.652	−0.574	−0.561	−0.615
	0.50, 0.001	1.00, 0.017	1.50, 0.055	2.00, 0.001	2.50, 0.044	3.00, 0.002	3.50, 0.065	4.00, 0.083	4.50, 0.006
L15	−0.008	(+0.056)	−0.321	−0.357	−0.260	−0.217	−0.217	−0.189	−0.206
model	−0.008	+0.044	−0.254	−0.353	−0.239	−0.213	−0.155	−0.186	−0.177
	0.50, 0.000	1.00, 0.013	1.50, 0.156	2.00, 0.004	2.50, 0.022	3.00, 0.004	3.50, 0.062	4.00, 0.003	4.50, 0.061
L21	−0.235	−0.247	−0.658	−0.664	−0.662	−0.661	−0.651	−0.665	−0.646
model	−0.203	−0.239	−0.470	−0.660	−0.655	−0.656	−0.647	−0.662	−0.646
	0.33, 0.032	0.67, 0.028	1.00, 0.187	1.33, 0.004	1.67, 0.007	2.00, 0.005	2.33, 0.005	2.67, 0.003	3.00, 0.007
L28	−0.002	(+0.054)	−0.063	(−0.042)	−0.160	−0.332	−0.417	−0.478	−0.446
model	−0.002	+0.056	−0.050	−0.042	−0.140	−0.327	−0.413	−0.448	−0.444
	0.25, 0.001	0.50, 0.003	0.75, 0.036	1.00, 0.001	1.25, 0.093	1.50, 0.042	1.75, 0.009	2.00, 0.032	2.25, 0.015
L35	−0.012	(+0.007)	(−0.006)	(+0.027)	(−0.021)	−0.076	(−0.109)	−0.257	−0.284
model	−0.012	+0.000	−0.006	+0.024	−0.031	−0.088	−0.078	−0.073	−0.261
	0.17, 0.000	0.33, 0.009	0.50, 0.001	0.67, 0.007	0.83, 0.019	1.00, 0.016	1.17, 0.031	1.33, 0.184	1.50, 0.025

is only 23.0% travel and therefore predominantly crane handling, whereas at the energy-optimal end it is 89.9% travel, a migration of -0.669 against the cell mean of -0.662 in Table 4.5.

The mechanism follows from the speed selection. Minimising the makespan drives every leg to the fastest available speed, which compresses travel until the crane chains, whose handling times are fixed and independent of vehicle speed, become the binding resource. Minimising energy does the reverse, selecting slow speeds that lengthen every leg until travel dominates the path again. The cranes therefore bind at the fast end and the vehicles at the slow end, which is why the transfer requires enough vehicles for the cranes to be competitive in the first place, and why it vanishes below $A/Q \approx 1$.

The single instance exhibiting the opposite sign, L01 at $A/Q = 1.00$ with $+0.108$ inverted relative to its neighbours, is the four-task instance, whose 35-point front offers too few structurally distinct schedules for the argument to apply.

A second campaign, independent of the sweep, holds each of the 35 published small instances at the fleet ratio its own configuration specifies and runs ten seeds of each. Table 4.6 groups its 350 replicates by ratio. The proportion whose migration exceeds 0.1 in magnitude rises monotonically from none at $A/Q = 0.33$ to 56% at $A/Q = 1.00$. Because the published configurations reach only $A/Q = 1.00$, this campaign observes the sub-threshold side alone, and it corroborates the ratio dependence on the whole set rather than on the six instances of the sweep. It also shows that the onset is gradual from about $A/Q = 0.4$ rather than a step at unity.

Several cells report a standard deviation of exactly zero across ten seeds, notably L01 throughout and L21 at the lowest ratio. In those cells every seed recovers the same two extreme schedules,

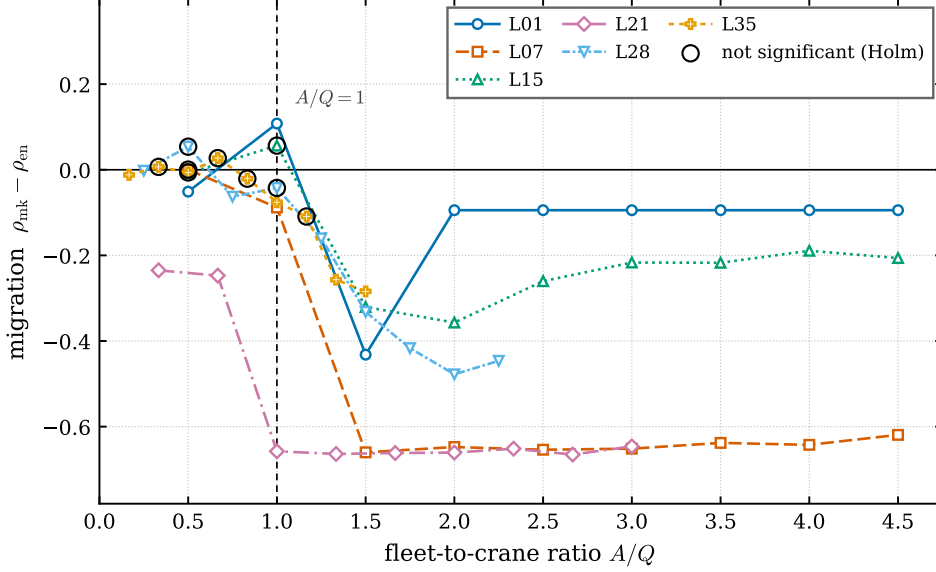


Figure 4.3: Migration against the fleet-to-crane ratio for the six swept instances, each cell the mean over ten seeds. Series are distinguished by marker and dash pattern so that they remain separable in greyscale. Open circles mark cells that fail significance after Holm correction; they are exactly the near-zero cells at low A/Q . The dashed vertical is $A/Q = 1$. The threshold structure is the result: migration is absent below $A/Q \approx 1$, grows sharply thereafter, and saturates. The crossing point is instance-dependent. The ratio ranges differ by instance because nine fleet sizes are swept against a fixed crane count that differs between them. Dispersion is omitted here and reported per cell in Table 4.5.

Table 4.6: Migration on all 35 published small instances at their native fleet ratios, ten seeds each, grouped by ratio. This campaign is independent of the sweep of Table 4.5, which varies the ratio on six instances; here the ratio is whatever the published configuration specifies. The proportion of replicates whose migration exceeds 0.1 in magnitude rises monotonically with the ratio, which corroborates the dependence on a larger set of instances. The ratios reach only 1.00, so the campaign probes the sub-threshold side alone. The decomposition closes on all 350 replicates, so none is excluded.

A/Q	n	mean migration	mean migration	migration > 0.1
0.33	10	+0.007	0.036	0%
0.40	40	+0.025	0.049	12%
0.50	60	-0.010	0.062	13%
0.67	80	-0.071	0.126	41%
1.00	160	-0.110	0.172	56%
Overall: $n = 350$, mean -0.065 , mean migration 0.124				

which is unsurprising for a four-task instance and indicates saturation of the search rather than of the measure.

4.5 The three instance families

Every result to this point rests on the loading family: in the 35 published small instances each of the 368 container moves is a loading task, so the unloading recurrence (2.4) is never exercised, and the dual-cycling instances of Section 4.3 are observed at a single fleet ratio. The source supplies the remedy without further data. Loading instances are Table 5 as printed; unloading instances are obtained from the same table by reversing the origins and destinations, so that a container originates at the crane and is delivered to the yard station; and a mixed instance is formed by combining one loading instance with one unloading instance. Applying these two constructions to Table 5 yields 35 unloading and 26 mixed instances alongside the 35 loading instances already used, on the same layout and with no value retyped.

Table 4.7: Migration by fleet-to-crane ratio on the three instance families. Loading is Table 5 as printed; unloading is the same data with origins and destinations reversed; mixed combines one loading with one unloading instance, following the construction stated in the source. Each cell is the mean over all instances of that family and ten seeds, restricted to the ratios every family attains. The threshold is reproduced independently on all three: migration is negligible below $A/Q \approx 1$ and saturates near -0.5 above it. Family sizes: loading 3,150 replicates and 232,835 front points; unloading 3,150 replicates and 230,211 front points; mixed 2,340 replicates and 194,853 front points.

A/Q	loading	unloading	mixed
0.25	-0.021	-0.014	-0.019
0.33	-0.043	-0.042	+0.205
0.50	-0.049	-0.022	+0.095
0.67	-0.060	-0.023	+0.075
0.75	-0.083	-0.017	-0.115
1.00	-0.106	-0.017	-0.153
1.25	-0.279	-0.166	-0.545
1.33	-0.359	-0.207	-0.197
1.50	-0.453	-0.291	-0.386
1.67	-0.527	-0.372	+0.005
1.75	-0.490	-0.400	-0.562
2.00	-0.497	-0.469	-0.410
2.25	-0.514	-0.448	-0.555
2.33	-0.538	-0.538	-0.179
2.50	-0.488	-0.483	-0.342
2.67	-0.550	-0.530	-0.179
3.00	-0.503	-0.480	-0.409
3.50	-0.500	-0.447	-0.407
4.00	-0.495	-0.445	-0.412
4.50	-0.473	-0.445	-0.409

The pairing that forms the mixed set is stated in the source: the “comb. of” column of Table 3 names, for each of the 26 instances, the unloading and loading instance it combines. That column is used as printed, so the family reproduces the published set rather than approximating it. It pairs by neither adjacency nor position, D04 combining U01 with L02 and D06 combining U02 with L01, and the reconstruction is confirmed against an independent published quantity: the task count of all 26 agrees with the Q-T column of the same table.

Each of the 96 instances was run at nine fleet sizes with ten seeds, and the transport share was computed at every point of every resulting front rather than at the two extremes alone. The campaign comprises 8,640 replicates and 657,899 front points; no replicate failed to complete and the decomposition closes at every point. Table 4.7 reports migration by ratio for each family and Figure 4.4 plots it.

The threshold established on loading instances therefore holds on all three families. Two differences are visible and neither follows from the mechanism of Section 4.4. Above the threshold the unloading family migrates no further than the loading family at any ratio, the gap running from 0.09 to 0.16 between $A/Q = 1.25$ and 1.75 and closing to zero at 2.33 ; the recurrences differ in where the handling term enters the maximum, which is a candidate explanation that was not tested. Below the threshold the mixed family carries a positive migration of up to $+0.205$ where the single-cycling families sit near zero, so on a mixed instance with few vehicles the makespan-optimal end is the more travel-heavy of the two. Both are reported as measured regularities and neither is claimed as understood.

Because the share is computed at every front point rather than at the two ends, the explanation can also be read as a profile along the front. Figure 4.5 shows one front from each family. The

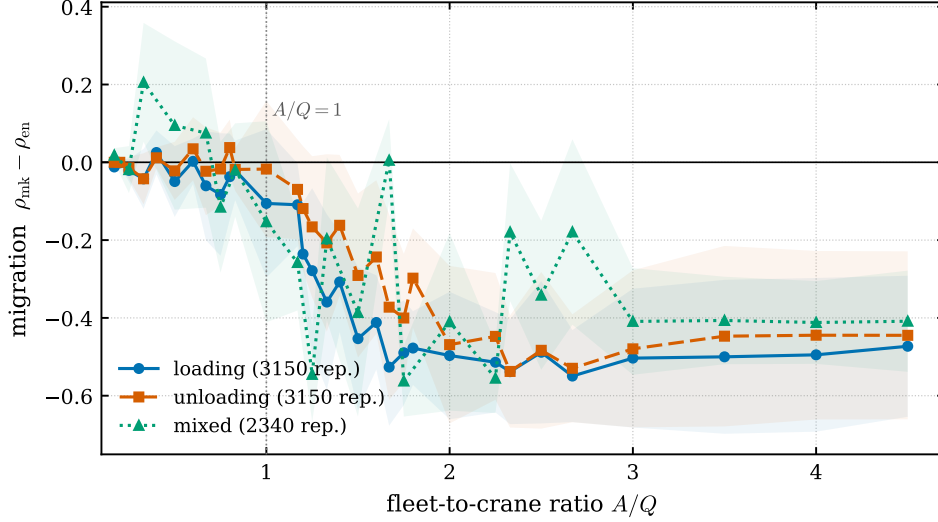


Figure 4.4: Migration against the fleet-to-crane ratio on the three instance families, with the shaded band one standard deviation across instances and seeds. The threshold is reproduced independently on each: migration is negligible below $A/Q \approx 1$, turns sharply negative through 1.2 to 1.5, and saturates near -0.5 . The mixed family departs from the other two below the threshold, where its migration is consistently positive rather than near zero.

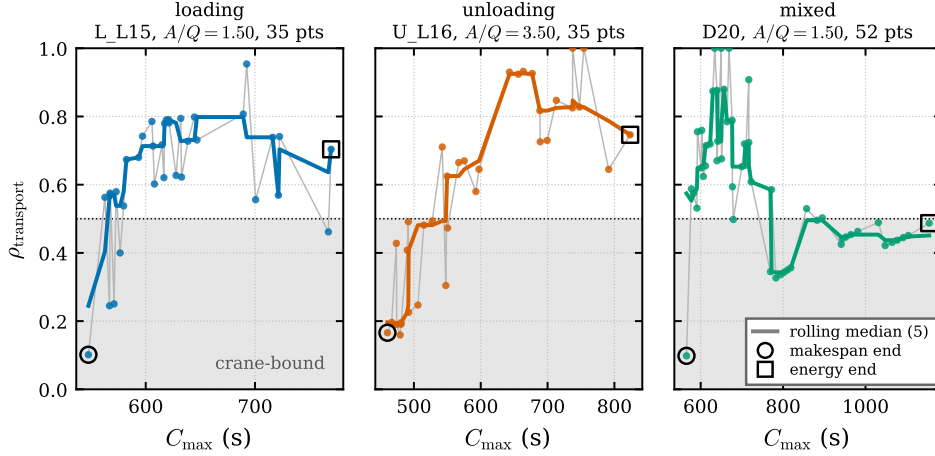


Figure 4.5: The transport share at every point of one front from each family, plotted against the makespan of that point. The dotted rule at $\rho_{\text{transport}} = 0.5$ separates the crane-bound from the vehicle-bound regime. The share rises from the makespan-optimal end to the energy-optimal end in all three, but not monotonically.

transition from a crane-bound to a vehicle-bound critical path occurs across the front rather than at its ends, and it is not monotone: neighbouring points differ in which resource binds, with the share moving by more than 0.3 between adjacent compromises. A two-point measurement reports the endpoints of this profile and is silent about its shape. Whether the non-monotonicity reflects genuine structure or the search returning structurally distinct schedules at neighbouring objective values is not resolved here.

4.6 How the explanation changes during the search

Every result to this point explains *finalised* solutions. Two questions remain open, and they are the two roles the surrogate can take beyond post-hoc explanation. Whether the explanation of a front is settled once the front is, or still moving while the search converges; and whether the learned model tracks that movement, which is what any later use of it inside the search would require.

Table 4.8: Surrogate calibration and post-hoc faithfulness over the sweep (mean over all fleet sizes and seeds per instance). These quantities belong to the secondary, over-requirements tier: they measure how well the model reproduces an explanation that the exact oracle already provides without it.

Instance	n	$R^2(C_{\max})$	$R^2(E)$	Leg Jaccard	$ \Delta\rho $
L01	90	0.9991	1.0000	0.990	0.0022
L07	90	0.9993	1.0000	0.882	0.0304
L15	90	0.9995	1.0000	0.917	0.0362
L21	90	0.9994	1.0000	0.885	0.0309
L28	90	0.9994	1.0000	0.888	0.0258
L35	90	0.9993	1.0000	0.825	0.0326

Both are answered by snapshotting the solver’s archive every ten generations. Six instances from each of the three families are run at three fleet sizes with five seeds, giving 54 cells, 270 replicates and 2,970 snapshots. At every snapshot the transport share is computed twice: once from the exact recurrence, and once from the surrogate, whose gradients are weighted by its own predicted durations so that the two explanations are never mixed. One model is fitted per instance and fleet size and reused across seeds, since it depends on the instance and not on the random stream. No replicate failed.

Figure 4.6 reports the first. The explanation is not settled early: above the threshold the mean migration moves from -0.101 at generation zero to -0.274 at generation one hundred, reaching its final value to within 0.02 only by generation forty; below the threshold it moves from -0.009 to -0.048 and settles by generation thirty. An explanation read from an unconverged front therefore understates the effect by more than half, because the early archive holds few structurally distinct schedules and its two extremes lie close together. This is a caution rather than a defect: it establishes that explaining *finalised* solutions, which is what Section 4.4 does, is the defensible use, and it gives the feedback loop of Chapter 5 a precondition, since guidance drawn from an early generation would act on an explanation that has not converged.

The surrogate tracks that movement closely. Over all 2,970 snapshots the median absolute difference between the model’s migration and the exact value is 0.0026, and the model is therefore reproducing the explanation while the search is still running, not only at its end. The agreement is family-dependent and its error distribution is skewed: the median is 0.0014 on loading, 0.0052 on unloading and 0.0062 on mixed instances, but the corresponding means are 0.022, 0.052 and 0.076, and the ninetieth percentile reaches 0.347 on the mixed family. The model is thus accurate in the typical case and occasionally far from it, which is the same pattern Section 4.7 reports for finalised solutions and has the same cause: where several resources are nearly balanced, a small error in a predicted duration changes which activity binds.

A second consequence concerns the generation budget. Chapter 5 notes that all runs use 100 generations against the published 300; the measurements here show migration settled well before the hundredth generation on both sides of the threshold, so the reduced budget is sufficient for this particular measurement, whatever it costs in absolute front quality.

4.7 Reproduction by the surrogate

Table 4.8 reports the secondary tier: whether the learned model reproduces the explanation that the max-plus layer supplies exactly.

Calibration is high, with R^2 on the makespan at 0.999 or above. The energy figure should not

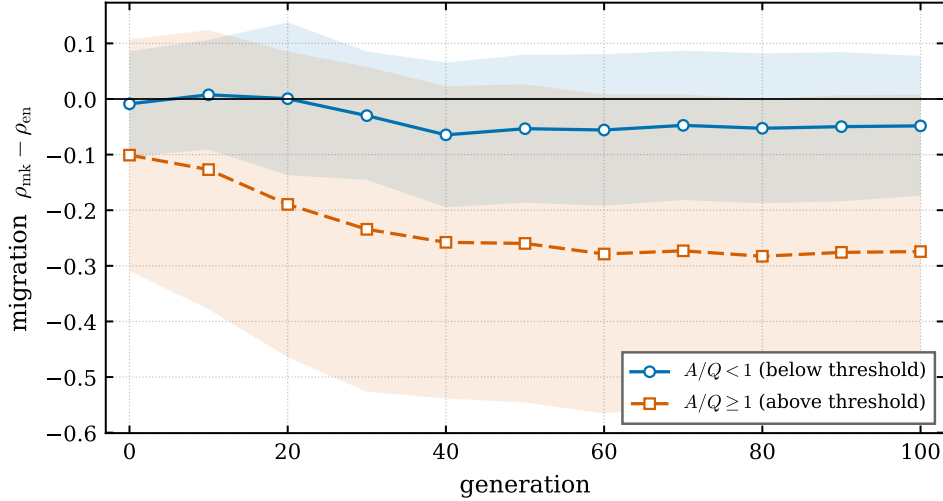


Figure 4.6: Migration against generation, over 270 replicates with the solver’s archive snapshotted every ten generations, grouped by side of the fleet-to-crane threshold with the band one standard deviation over replicates. Above the threshold the explanation moves from -0.101 to -0.274 and reaches its final value to within 0.02 only by generation forty; below it there is little to settle. An explanation read from an unconverged front is therefore biased towards zero.

be read as a success, for the architectural reason given in Section 3.4.

The meaningful quantity is the reproduction of ρ . The mean absolute difference between the surrogate’s migration estimate and the exact value is 0.026 across the sweep, against migration magnitudes reaching 0.66 , a relative error of approximately five per cent on the effect being measured. On the cells of largest migration the discrepancy is also largest, reaching 0.187 on L21 at $A/Q = 1.00$, so the reproduction is weakest precisely where the phenomenon is strongest. That anti-correlation is the honest summary of the secondary result, and it is the reason the two tiers are reported separately throughout. The exact explanation of Tables 4.3–4.5 does not degrade when the surrogate does.

4.8 Discussion

The results support a specific and bounded claim. For this problem the reason a solution is Pareto-optimal can be stated exactly, in seconds attributable to named physical activities, with the attribution summing to the objective by construction. This is stronger than a feature attribution over decision variables, and it is available because the objective possesses $(\max, +)$ structure that the max-plus layer preserves rather than because of any technique imported from machine learning. The architectural lesson is narrow but transferable: the encoder was not the obstacle to faithful attribution, the pooled regression head was, and replacing that head with the problem’s own recurrence removes the obstacle without sacrificing the learned representation.

The explanation is not static along the front. Where vehicles are sufficiently plentiful, trading time for energy transfers the binding constraint from the cranes to the vehicles by an amount reaching two-thirds of the critical path; where they are scarce, as in every published large instance, it does not, there being no second candidate for the bottleneck. An operator reading a front therefore cannot assume its members are limited by the same resource, and A/Q indicates which regime obtains. The surrogate reproduces that explanation to a mean absolute ρ error of 0.026 , with critical-leg agreement of 0.898 on the small instances and 0.924 on the published large

ones, and it does so least accurately where the effect is largest, since Pareto-optimal schedules are those in which resources are nearly balanced. The exact recurrence remains the yardstick, but it can only be applied to a schedule that has already been evaluated; the model predicts the durations and can therefore explain a candidate before that evaluation exists, which is the setting the feedback loop of Chapter 5 would require.

Limitations and Future Work

Ties in the argmax. An earlier version of the attribution produced, on a small fraction of schedules, a critical path whose durations did not sum to the makespan. The faulty component was not the tropical layer but the reduction after it: the makespan is a maximum over several terminal nodes, and that reduction used the framework’s own maximum operator, which divides the incoming gradient equally between tied maxima, so each activity on both paths inherited a fractional share and the strict on-path test admitted neither. Any single maximiser is a valid subgradient of a maximum; an average of several is not a valid selection for an additive decomposition. The correction selects the maximal terminal by index, restoring the convention the recurrence already follows, and was applied at every terminal reduction including the batched form. Across 43 non-closing schedules from seven instances of all three families every activity gradient was at most 0.5, so the tie accounts for every case examined. Every affected computation was repeated from its recorded seed: the decomposition now closes on all 940 replicates and at all 657,899 front points, against nine replicates and 1.30% of points before, with conclusions unchanged and only one of 54 cells moving at all, by 0.001. Why the pre-correction failure rate differed by family, at 0.16% on loading against 2.94% on unloading, was not established.

Scope of the claim. All runs use 100 generations against the published 300, which suffices for a measurement depending on the two front extremes but places absolute front quality below what the method attains, so no claim about front quality should be drawn here. The sweep establishes that migration depends on A/Q and locates the transition between 1.0 and 1.35 for four instances, but does not predict it from instance parameters. It is also established on the small-instance layout alone: the large dual-cycling instances describe a twenty-crane terminal whose distance matrix is not interchangeable, and are observed only at $A/Q = 0.50$, so extending the threshold to them is an extrapolation across layouts rather than a measurement. Sweeping the fleet on three large dual-cycling instances would settle it, at roughly forty core-hours. Finally, the surrogate’s agreement with the exact critical set is lowest where migration is largest, since Pareto-optimal schedules are those in which resources are nearly balanced and small prediction errors alter which activity binds; this bounds the secondary tier only.

Further work. Two implemented components are excluded. A power-coupled regime, in which a terminal-wide budget couples the vehicles, is the one setting in which the surrogate would predict a quantity with no closed form, the waiting time induced by power contention, and therefore where a learned component becomes necessary rather than merely sufficient; it is the strongest candidate for future work but needs a separate problem statement, originating as it does in a different literature. Its three-velocity configuration and power coefficients are retained here, belonging to the original problem. The natural continuation is to feed the explanation

back into the search, which divides into two steps of unequal size. The smaller, comparing how the explanation changes across generations, is carried out in Section 4.6 and shows that an explanation read from an unconverged front is biased towards zero. The larger is attribution-guided search proper, biasing mutation towards the activities currently on the critical path, in the spirit of decision-focused learning [8]. That step is deliberately not attempted here: it should follow once the explanation is established, which is what this thesis does, and Section 4.6 gives it a precondition, since guidance drawn from an early generation would be acting on an explanation that has not yet converged.

Chapter 6

Conclusion

This thesis set out to explain why particular solutions to a bi-objective container-terminal scheduling problem are Pareto-optimal, rather than merely to compute them. Its one core idea is that such a schedule can be explained exactly, in seconds attributable to named physical activities, and that a learned model can produce the same explanation. The instances are the published ones throughout: the loading set as printed, the unloading set reversed from it, and the mixed set formed by the pairing the source states, each reconstruction checked against the optima published for it before any result was drawn from it.

Because the makespan is a maximum of sums of activity durations, it is a longest path over the $(\max, +)$ semiring, and the sensitivity of the makespan to each activity is exactly zero or one. A conventional surrogate cannot express this: a pooled regression head produces a dense Jacobian and therefore assigns credit to provably irrelevant activities. Replacing that head with a differentiable max-plus dynamic programme, whose backward pass routes the subgradient through the recorded argmax edge, restores the property, and differentiating the composition returns the critical path itself rather than an estimate of it. From that decomposition the duration-weighted critical share ρ summarises an explanation as one number, the fraction of the makespan spent moving rather than lifting.

The empirical result is that the bottleneck migrates, and that whether it does is governed by the fleet-to-crane ratio. Across 540 statistically tested replicates migration is significantly non-zero in 45 of 54 instance–fleet cells and attains a magnitude of 0.665; below $A/Q \approx 1$ it vanishes, both front extremes being transport-bound. A further 8,640 replicates reproduce the threshold on three instance families, so the result does not depend on every container being a loading move. Where it occurs the cranes bind at the makespan-optimal end and the vehicles at the energy-optimal end, which accounts for its absence on the ten published large instances, each having $A/Q = 0.50$. The operational reading is that two schedules on one front need not be limited by the same resource, and that the fleet ratio indicates whether an operator is in the regime where this distinction matters.

The learned surrogate reproduces the same explanation to a mean absolute ρ error of 0.026, with critical-leg agreement of 0.898 on the small instances and 0.924 on the largest published ones at up to 160 tasks. That is what licenses the method beyond the cases examined here: the exact recurrence is available for a resolved schedule, but it requires one, whereas the model predicts the durations and can therefore explain a candidate that has not been evaluated. Every setting in which the explanation is wanted before evaluation, or in which no closed form exists, rests on the learned component rather than on the oracle.

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